

Uncertainty

JEB104 Microeconomics I
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Do You Like Risk?

Choice task:

- Lottery with 50% chance of winning \$300 and 50% chance of getting zero
- Getting given amount for sure (gradually increasing)

When do you start choosing the safe payment instead of lottery?

When do you think other people start choosing the safe payment?

Previously

- Everything was certain
- We analyzed choices people make when they know exactly what will happen
- We have a quite complete theory of demand and of how prices affect decisions

Now: Introduce uncertainty

- How do people make decisions when they are not certain what will happen?
- Uncertainty involves risk, every choice gives uncertain outcome.

Imagine that you work and you have two alternative specifications for your reward:

- Flip a coin and then get \$300 or nothing (lottery with 50% chance of winning \$300 and 50% chance of getting zero)
- get X for sure

Which one would you choose? It will surely depend on X .

- If $X = 10$, then probably the lottery.
- If $X = 290$, then probably the sure amount.

Do You Like Risk?

#	Safe payment	Lottery
1	10	50 % getting 300 and 50 % getting 0
2	20	50 % getting 300 and 50 % getting 0
3	30	50 % getting 300 and 50 % getting 0
4	40	50 % getting 300 and 50 % getting 0
5	50	50 % getting 300 and 50 % getting 0
6	60	50 % getting 300 and 50 % getting 0
7	70	50 % getting 300 and 50 % getting 0
8	80	50 % getting 300 and 50 % getting 0
9	90	50 % getting 300 and 50 % getting 0
10	100	50 % getting 300 and 50 % getting 0
11	110	50 % getting 300 and 50 % getting 0
12	120	50 % getting 300 and 50 % getting 0
13	130	50 % getting 300 and 50 % getting 0
14	140	50 % getting 300 and 50 % getting 0
15	150	50 % getting 300 and 50 % getting 0
16	160	50 % getting 300 and 50 % getting 0
17	170	50 % getting 300 and 50 % getting 0
18	180	50 % getting 300 and 50 % getting 0
19	190	50 % getting 300 and 50 % getting 0
20	200	50 % getting 300 and 50 % getting 0

Uncertainty

Events and contingencies are often uncertain

How are choices made under uncertainty?

Attitudes toward risk:

- risk averse
- risk loving

What can people do to avoid or reduce risk?

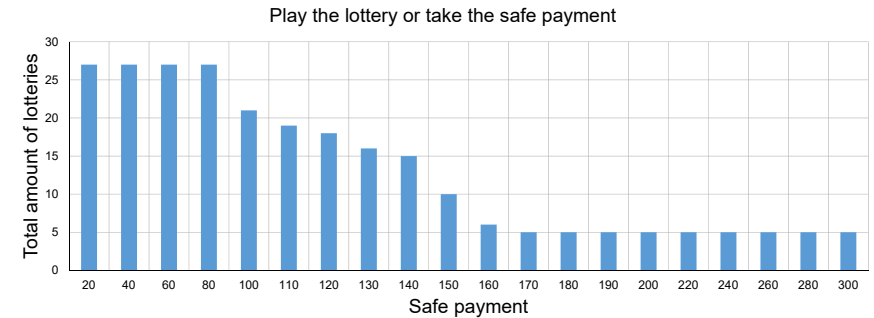
- insurance
- risk diversification

For which sure amount would you start to prefer it to playing lottery?

You may also imagine that you own your lottery ticket and you know that if you play the lottery you can get 300 dollars with 50% chance and 0 otherwise.

You have an option to sell this lottery ticket. For which amount would you be willing to sell the ticket and get the amount for sure?

Data generated by students last year – 27 students participated:



For safe payment ≥ 170 , everybody (except 5 lovers of lotteries) decided to rather take the safe payment than to play the lottery.

So far, we have assumed that people's choices do not involve any degree of uncertainty; once they decide what to do, they get what they have chosen. That is not always the way things work in many real-world situations.

Uncertainty present in many real-world situations.

- Sometimes you do not know what kind of goods you are buying (rotten apples).
- Buying a lottery ticket, investing in shares of common stock, or playing poker involves a lot of uncertainty.
- Even interest rates are not known on e.g. long-term mortgages or on your savings.

Other examples of contingent uncertainties:

- Holidays in Asian country, but then tsunami hits just before departure.
- Lightning/flooding can destroy your house.
- Sunny morning, so you go on a bike ride, after you leave it starts raining.

Different attitudes toward risk

- Some people do not like uncertainty and are scared when undertaking risky activity (expect that the outcome will be the bad one).
- Some other people like to take risky activities, because the fact that something is risky increases their adrenalin.

Reducing risk (discussed in the next lecture)

- You can insure your holiday package $\Rightarrow$ if you can't go (got injured, tsunami) you will not lose all your money.
- House insurance $\Rightarrow$ pay the insurance company, if lightning/floods strike, you get some money for rebuilding the house.
- Investing into assets with different risk.

Probability Distribution

Probability

- the relative frequency with which an event occurs

Probability distribution

- likelihood that a given random variable will take on any given value

Discrete probability distribution

- finite number of events

Example:

- people can be 160cm, 170cm, or 180cm tall.
- heights are spread equally throughout the hypothetical population.

Now we will go through very general and easy introduction to statistics that will be necessary in further analysis. Maybe that I only repeat what you have already learned.

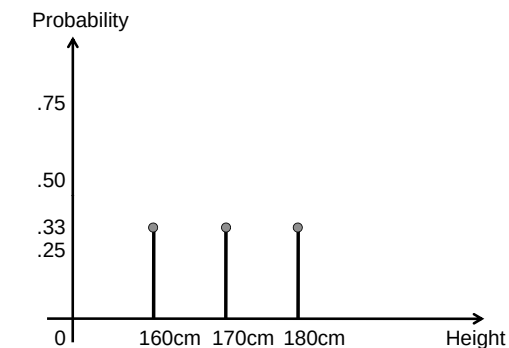
First example: The probability of a head coming up on the flip of a fair coin is $1/2$

- That means that if a coin is flipped a large number of times, we can expect a head to come up in approximately one-half of the flips.
- Of course, before a coin is flipped, we have no idea what will happen, so each flip has an uncertain outcome (it is a random variable).

Height of people:

- Suppose we sit in a room and measure the height of each person who enters that room, the height of any entrant will be a random event.
- It may be 160 cm, 170 cm, or 180 (well, if we are not at the meeting of basketball players).
- But if we know how height is distributed in population, we can place a probability that a person coming will be of any particular height.

Discrete Probability Distribution



What is the expected height of a person randomly drawn from the hypothetical population?

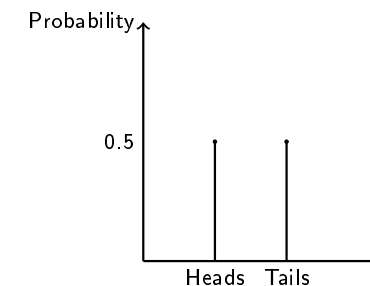
$$\text{Expected height} = \frac{1}{3} \cdot 160\text{cm} + \frac{1}{3} \cdot 170\text{cm} + \frac{1}{3} \cdot 180\text{cm} = 170\text{cm}$$

As heights are spread equally throughout the population, the probability that a person coming will be 160cm tall is $1/3$. This is also the probability that a person coming will be 170cm (180cm) tall.

The figure shows the discrete probability distribution of height, each possible height has the probability 0.33.

To get expected height, we just multiply values by their probabilities and sum them (mathematically, we compute a weighted average, and weights are probabilities.)

Coin flipping – the probability distribution, options heads and tails with prob. 0.5



Discrete Probability Distribution

Expected value (mean):

$$\bar{x} = \sum_i \pi_i x_i$$

π_i – the probability of event i occurring

x_i – the value of event i when it occurs

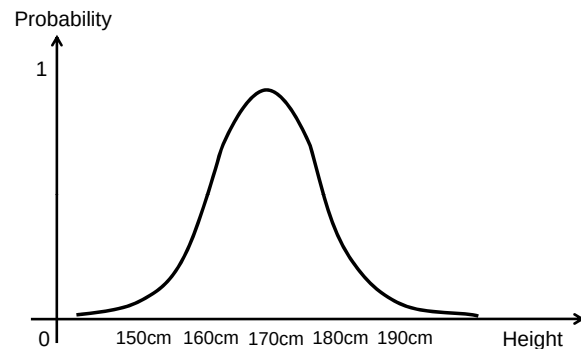
Two conditions on probabilities:

- 1 $1 \geq \pi_i \geq 0$
- 2 $\sum \pi_i = 1$

Continuous Probability Distribution

Continuous probability distribution

- infinite number of events



$\sum \pi_i = 1$ means that some event must happen, so probabilities have to sum to 1

In reality, we however have many options of heights of people.

- Distribution of height is more likely to be continuous.

Shape

- Tiny probability that a person would measure 140cm or 210cm
- People mostly measure around 170cm

The curve envelops vertical segments that could be drawn from discrete distribution.

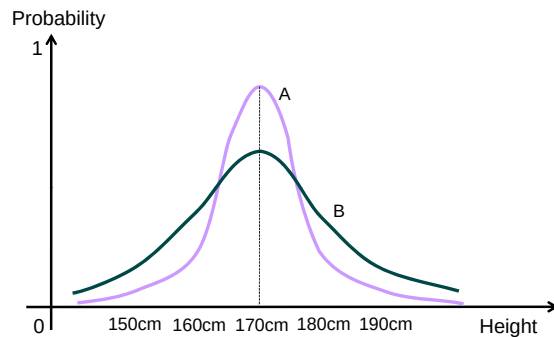
Please note that the value 1 on the vertical axis should be much further.

The figure shows a symmetrical distribution – cut in the middle and the left part could be mirrored to get the right part.

Expected values of such symmetrical distributions are exactly in the middle.

Properties of Probability Distribution

Symmetrical distribution: mean $\bar{x}$ in the middle



Distributions A and B have the same mean.

A “skinnier” than B → higher chance that event can take extremely high or low values for B → more variability in B

Variance: $\sigma^2 = \sum \pi_i (x_i - \bar{x})^2$, $\sigma_B^2 > \sigma_A^2$

The figure shows two symmetrical distributions that have the same expected value (mean). However, they look differently. How can we describe this difference?

We can use variance. So each distribution can be described by its mean and variance.

Definition of variance:

- average distance of random numbers from the mean weighted by probabilities of these random numbers
- it measures how far a set of (random) numbers are spread out from their average value

The higher variance, the more numbers are more “spread”, so there is a greater variability of the random numbers that occur.

Properties of Probability Distribution

Assume we have n random variables: $x_1, x_2, \dots, x_n$

- all x_i have the same probability distribution
- mean $\bar{x}$, variance σ^2

Construct new random variable y : $y = \frac{1}{n}(x_1 + x_2 + \dots + x_n)$

- mean $\bar{y} = \bar{x}$
- variance $\sigma_y^2 = \frac{1}{n}(\sigma^2)$

This comes from statistical properties and definitions of mean and variance.

- The new random variable is an average of the original random variables.
- Mean is the same, but variance is much lower.
- In simple words, due to squared form of variance, you multiply by n but divide by n^2 .

The example follows on the next slide.

Properties of Probability Distribution: Diversification Example

Investment to Czech and Slovak funds

- Both are random variables
- Each 1 dollar invested yields with 0.5 probability 2 dollars and with 0.5 probability 0 dollars (for each fund)
- You have 100 dollars to invest

Investing 100 dollars in one of the two funds:

- $\bar{x} = 0.5 \cdot 200 + 0.5 \cdot 0 = 100$
- $\sigma^2 = 0.5(200 - 100)^2 + 0.5(0 - 100)^2 = 10,000$

Investing 50 dollars to each fund:

- $\bar{y} = 0.25 \cdot 0 + 0.25 \cdot 100 + 0.25 \cdot 100 + 0.25 \cdot 200 = 100$
- $\sigma_y^2 = 0.25(0 - 100)^2 + 0.5(100 - 100)^2 + 0.25(200 - 100)^2 = 5,000$

Imagine that you are considering investment to the Czech and Slovak fund.

Each fund has same payouts and probabilities, so probability distribution is the same

$E(x) = 100$, cost = 100 $\Rightarrow$ Obviously this would be a bad investment, but this is an illustrative example only

Investing into one fund:

- you either earn extra \$100 (giving you \$200 in total)
- or you earn nothing and losing your \$100

Investing into both funds results in 4 options (which stack into three at the end):

- getting 0 (nothing from both funds) – probability 0.25
- 100 (Slovak fund is the only one earning) – probability 0.25
- 100 (Czech fund is the only one earning) – probability 0.25
- 200 (both funds are earning) – probability 0.25

Investment into both funds has the same expected mean as investment into one fund, but lower variance, so it reduces risk.

This is important for portfolio diversification

- Will be covered in detail in finance classes

In the next part, we will talk only about monetary gambles.

Decision-maker chooses between investments A and B, does not know outcome of investments, because the outcome depends on weather.

- Very hot and wet weather is the most favorable for planting rice as it yields the highest payoff.
- Different geographical areas have different weather.
- Investments in different areas give different expected payoffs.

State-contingent means depending on something not yet certain, i.e. taking into account every possible option (it is a specific weather in this example) that may occur.

Decision Making under Uncertainty

Two risky investments involving rice production in different geographical areas

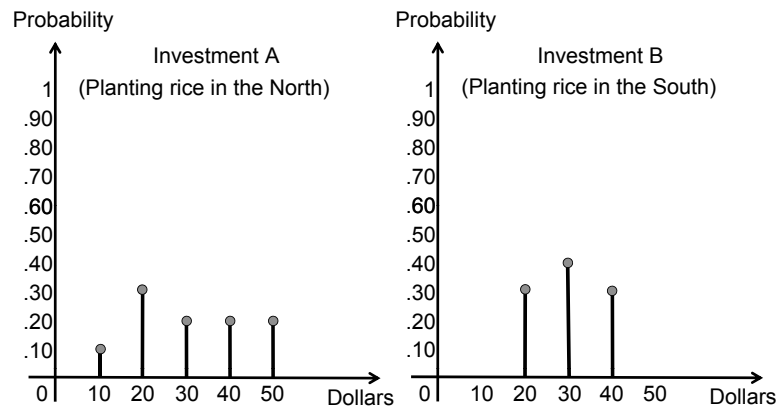
5 states of nature (weather): (i) cold and dry, (ii) cold and wet, (iii) hot and dry, (iv) hot and wet, (v) very hot and wet

State-contingent outcomes:

Weather	Investment A Plant rice in the north		Investment B Plant rice in the south	
	Payoff	Probability	Payoff	Probability
Cold and dry	\$10	0.10	\$10	0
Cold and wet	\$20	0.30	\$20	0.30
Hot and dry	\$30	0.20	\$30	0.40
Hot and wet	\$40	0.20	\$40	0.30
Very hot and wet	\$50	0.20	\$50	0

Decision Making under Uncertainty

Two investments in rice production



Decision Making under Uncertainty

This Figure is just a graphical illustration of probability distribution of the two investments from the previous slide.

Dollars on horizontal axis correspond to each specific weather condition.

Our farmer decides on the investment based on expected monetary return, so we calculate it for both investments A and B.

This decision takes only EV into account. But this is often unrealistic – see the next slide.

Decision based on the expected monetary value/return:

$$EV = \sum_i \pi_i X_i$$

= the average outcome from an uncertain gamble
 (weighted average of returns where weights are probabilities)

Investments to rice production give the following expected returns:

- $EV_A = 0.10 \cdot 10 + 0.30 \cdot 20 + 0.2 \cdot 30 + 0.2 \cdot 40 + 0.2 \cdot 50 = 31$
- $EV_B = 0 \cdot 10 + 0.30 \cdot 20 + 0.4 \cdot 30 + 0.3 \cdot 40 + 0 \cdot 50 = 30$
- In this case $EV_A > EV_B \Rightarrow$ agent chooses investment A

St. Petersburg Paradox

Should we really simply maximize expected monetary return?

St. Petersburg paradox

- Coin flipping until it lands heads up
- 50 % probability of landing heads up on any flip
- The coin lands heads up for the first time in n^{th} flip $\rightarrow$ payoff 2^n

$$EV = 2 \cdot \left(\frac{1}{2}\right) + 2^2 \cdot \left(\frac{1}{2}\right)^2 + 2^3 \cdot \left(\frac{1}{2}\right)^3 + \dots + 2^n \cdot \left(\frac{1}{2}\right)^n = 1 + 1 + 1 + \dots = \infty$$

How much is our player willing to pay to take part in the game?

- maximizing expected monetary value $\rightarrow$ willing to pay infinite amount

St. Petersburg Paradox: Solution

Solution of St. Petersburg paradox:

Daniel Bernoulli: “*The determination of the value of an item must not be based on the price, but rather on the utility it yields.*”

Utility function $u(x)$, diminishing marginal utility of money $u''(x) < 0$

Expected utility from the gamble converges to finite value:

$$EU = u(2) \cdot \left(\frac{1}{2}\right) + u(2^2) \cdot \left(\frac{1}{2}\right)^2 + u(2^3) \cdot \left(\frac{1}{2}\right)^3 + \dots + u(2^n) \cdot \left(\frac{1}{2}\right)^n$$

Simply maximizing the expected return could not be a right approach. Let's have a look at the game described as St. Petersburg's paradox (introduced by Nikolaus Bernoulli in 1713)

- If coin lands heads up in the first slip – payoff 2, if on the second flip – payoff 4, if on the third flip – payoff 8....
- Computation of expected value and probabilities:
 - Suppose there are only 2 rounds, probability that game ends in the 1st round is 0.5.
 - Probability, that we move to the 2nd round is 0.5 and then we flip coin again.
 - The probability that coin lands heads up in the second flip is 0.5, but the probability that we get to the second flip is 0.5 (only if coin lands heads up in the first flip), so joint probability that an individual gets payoff 2^2 (coin lands heads up in the second flip) is $0.5 \cdot 0.5 = 0.25$.

If you had to pay to play such a game, how much would you pay?

If you valued the gamble based on its expected value, you would be willing to pay any amount (infinity)!

Daniel Bernoulli

- 18th century Swiss mathematician and statistician
- Not the same guy that invented the St. Petersburg paradox (his brother or cousin)
- There were actually several mathematicians/statisticians named Bernoulli (parents and children)

Note that the negative second derivative means that the utility function is concave, e.g. every additional unit brings a lower level of satisfaction.

Instead of expected monetary return, we derive expected utility.

Maximizing Expected Utility

Expected utility hypothesis

- Agent chooses between different gambles
- Each gamble assessed according to the expected utility the agent gets
- Agent chooses the gamble yielding the highest expected utility

Expected utility:

$$EU = \sum_{i=1}^n \pi_i u(x_i)$$

EU : von-Neumann-Morgenstern expected utility function

$u(x_i)$: utility from payoff x_i (Bernoulli utility function)

π_i : probability of event i with payoff x_i

Note: not unique (preferences preserved by a positive affine transformation) $EU' = a \cdot EU + b$

Need for Cardinal Utility

Ordinal utility is NOT enough

Need to introduce cardinal utility (differences of utility values have meaning)

	$u_1(x)$	$u_2(x)$
Apple	70	4
Orange	50	2
Chocolate bar	100	5

Choice between:

- apple for sure
- 50 % chance of getting orange and 50 % chance of getting chocolate bar

Expected utility from the gamble:

$$EU_1 = 0.5 \cdot 50 + 0.5 \cdot 100 = 75, \quad EU_2 = 0.5 \cdot 2 + 0.5 \cdot 5 = 3.5$$

$u_1(x) \rightarrow$ gamble is played, $u_2(x) \rightarrow$ apple is chosen

Gamble

- It can be monetary gamble, investment (decide where to invest your money), or just simple decision whether to go for a bike ride or whether to stay at home (prob. that it will be raining).

von-Neumann-Morgenstern

- John von Neumann was one of the major figures in mathematics in the twentieth century. He also contributed several important insights to physics, computer science, and economic theory.
- Some call him “the most influential mathematician who ever lived” (if the influence of a scientist is interpreted broadly enough to include impact on fields beyond science proper)
- Oscar Morgenstern was an economist at Princeton who, along with von Neumann, helped to develop mathematical game theory.

So far we could work only with ordinal utility

- Only ranking of alternatives important, utility values had no important meaning.
- The analysis could be derived from indifference curves that are used to order alternatives.

Ordinal utility is NOT enough now.

- Once probabilities have a role in the decision making process, utility values assigned to different outcomes start to matter.

The table shows two different utility functions that assign utility values to different goods: apple, orange, chocolate bar.

- The ranking of alternatives is the same for both utility functions.
- We add uncertainty and compute expected values for the gamble (orange or chocolate bar with probability 0.5).
- When the expected utility of the gamble is compared with the utility from an apple, different utility functions result in different choices.
- So differences in utility levels are important now.

Risk Neutrality

“Fair” gamble (sure amount = EV, no extra value of playing the gamble):

- A: 50 dollars for sure
- B: 50 % chance of getting 100 dollars and 50 % chance of getting 0

Risk neutral agent is indifferent between A and B

Utility function of risk neutral agent is **linear**.

$$u(x) = ax, \quad a > 0$$

$u'(x) = a \rightarrow$ marginal utility of income is constant

Different people have different attitudes towards risk.

Fair gamble

- expected value of the gamble equals its (opportunity) cost
- Examples:
 - Ticket to the lottery (win \$ 100 with probability 0.5 and get nothing otherwise) costs \$50.
 - Get \$50 or play the lottery (win \$ 100 with probability 0.5 and get nothing otherwise).

Consider buying lottery ticket which costs \$ 50:

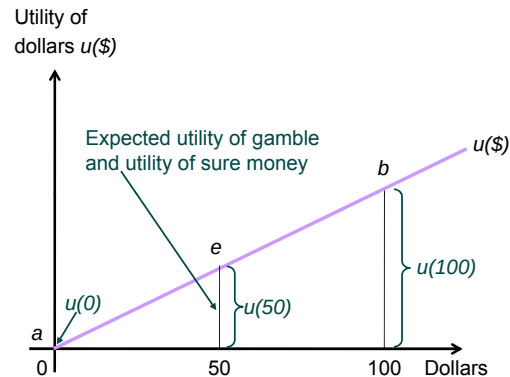
- Not to buy the lottery ticket: income m
- To buy the lottery ticket and earn EV: $m - 50 + 0.5 \cdot 100 + 0.5 \cdot 0 = m$

From observing agent's utility function we can get information about his attitude toward risk

Risk neutral individual – marginal utility of income is constant

- Each additional dollar has always the same value, no matter how high is the wealth

Risk Neutrality



Expected utility of the gamble: $EU = 0.5u(100) + 0.5u(0)$

Utility of the safe payment: $u(50) = u(0.5 \cdot 100 + 0.5 \cdot 0) = u(EV)$

$$EU = u(EV)$$

In general: $\pi u(L) + (1 - \pi)u(H) = u(\pi L + (1 - \pi)H)$, $L(H)$ low(high) outcome

Risk neutrality means that agent only cares about expected wealth.

$$u(50) = a \cdot 50 \text{ and } EU = 0.5 \cdot a \cdot 100 + 0.5 \cdot a \cdot 0 = 50a$$

Hence, $EU = u(EV)$.

Expected utility graphically:

- Connect utility levels $u(L)$ and $u(H)$ by a straight line.
- Notice that EU is a linear combination of these utility levels, so to determine EU , we need to find a point on this line that corresponds to the level of EU .
- Location of this point on the line is given by probabilities which are assigned to low and high outcome, if probability is 0.5 for both, the point will lie exactly in the middle. The higher is the probability of low outcome, the closer this point will lie to $u(L)$.
- Note that EV on the horizontal axis is also formed by using these probabilities ($EV = \pi \cdot L + (1 - \pi) \cdot H$).
- So if you draw a vertical line connecting EV on the horizontal axis and the straight line connecting $u(L)$ and $u(H)$, the length of this line graphically illustrates the size of EU .

For risk-neutral individual with linear utility function, the EU and $u(EV)$ overlap in the figure.

Risk Aversion

Most people are risk averse, when having large wealth, marginal utility of additional dollar is lower than if they are poor and their wealth is lower.

Ask yourself: Would you play the gamble giving \$100 with probability 0.5 or would you take the 50 dollars for sure?

- If you decide to take 50 dollars for sure, you are risk averse.

Risk averse agent prefers A to B and will reject to play the fair gamble.

Utility function of risk averse agent is **concave**.

Marginal utility of income is diminishing.

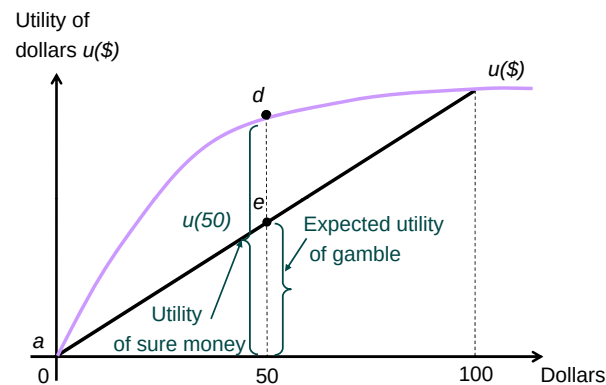
$$u(x), u'(x) > 0, u''(x) < 0$$

$$EU = \pi u(L) + (1 - \pi)u(H) < u(\pi L + (1 - \pi)H) = u(EV)$$

Risk Aversion

So if you ask a risk averse person, they will always choose EV for sure over the gamble.

EU is formed the same way as before. Because of concave utility function, utility of the expected value will be higher than the EU .



$$u(EV) = u(50) > EU = 0.5u(100) + 0.5u(0)$$

Risk Loving

Risk loving agent prefers B to A and will play the fair gamble.

Utility function of risk loving agent is **convex**.

Marginal utility of income is increasing.

$$u(x), u'(x) > 0, u''(x) > 0$$

$$EU = \pi u(L) + (1 - \pi)u(H) > u(\pi L + (1 - \pi)H) = u(EV)$$

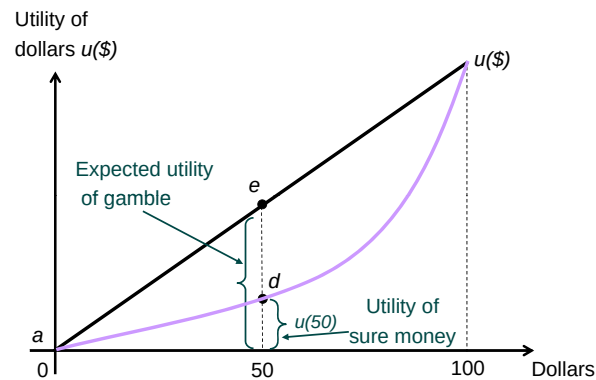
If you love risk, then the utility of the gamble is higher than its expected value.

- You get additional utility from being exposed to risk.

Risk Loving

In fact, a risk loving agent would be willing to pay to engage in this gamble.

However, people are mostly risk averse.



$$u(EV) = u(50) < EU = 0.5u(100) + 0.5u(0)$$

Do You Like Risk?

	Safe payment	Lottery
1	10	50 % getting 300 and 50 % getting 0
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Risk Aversion in Experiment

Now get back to our “experiment” from the beginning.

In which rows would you play lottery and when would you start taking the safe payment?

Risk loving agent switches decision later

- But notice that being risk loving agent only says that his expected utility is higher than utility of 150 for sure, but we don't know if it is also higher than his utility of e.g. 170 or 180.

People start with playing lottery

When do they switch to taking the safe payment?

Risk neutral agent:

- Safe payment < 150: chooses playing lottery
- Safe payment 150: indifferent
- Safe payment > 150: takes the safe payment
- switches decision on 15th or 16th row

Risk averse agent switches decision earlier

Risk loving agent switches decision later

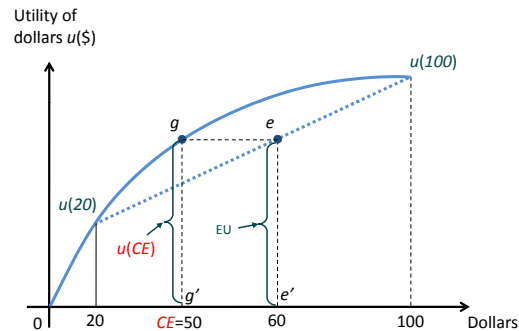
How to Quantify Degree of Risk Aversion

Certainty equivalent (CE)

- the lowest possible amount that someone is willing to take (for sure) in order not to participate in a gamble
- $u(CE) = EU$

Lower CE $\leftrightarrow$ higher risk aversion

Gamble: \$20 with probability 0.5 and \$100 with probability 0.5



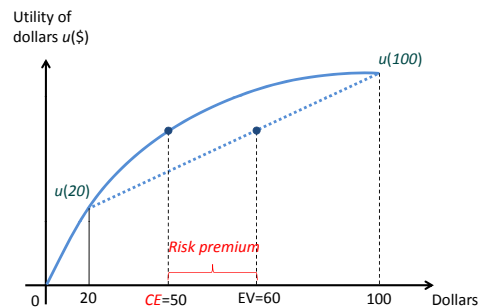
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How to Quantify Degree of Risk Aversion

Risk premium

- the difference between the expected value of a gamble and its CE
- positive for risk averse tastes, zero for risk neutral tastes and negative for risk loving tastes
- risk aversion: the amount consumer is willing to sacrifice in expected value in order to eliminate the risk she faces
- risk lover: the amount consumer must receive in addition to expected value in order to give up the chance to play the gamble



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How can we quantify degree of risk aversion?

- One way to do this is to begin by asking the following question: What is the least I have to give the person (for sure) in order for her to agree not to participate in the gamble at all?
- Graphically, the greater is the curvature of the utility function, the more risk averse individual is.

Why $u(CE) = EU$

- Because if $u(CE) > EU$ then you can lower down the CE and the person will still agree not to gamble
- And if $EU > u(CE)$ then it pays off to participate in the gamble
- So if $u(CE) = EU$ you are indifferent

Example: consider the following gamble

- With probability 0.2 you get 10,000
- With probability 0.8 you get 2,500
- utility function $u(m) = \sqrt{m}$
- $EU = 0.2\sqrt{10,000} + 0.8\sqrt{2,500} = 0.2 \cdot 100 + 0.8 \cdot 50 = 20 + 40 = 60$
- $u(CE) = EU \Rightarrow \sqrt{CE} = 60 \Rightarrow CE = 3600$

- Risk neutral individual: $EU = u(EV)$, so $CE = EV$
- Risk loving individual: $EU > u(EV)$, so $CE > EV$

So to put it in shorter terms, risk premium is the difference between the expected value of the gamble and its certainty equivalent:

$$RP = EV - CE$$